MTH208

Timed Online Assignment
(July 2020 Semester- Answer Keys):
For students' practice and reference

Question 1 (Total: 15 marks)

Calculate the Jordan canonical form of the matrix

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{pmatrix}.$$

(15 marks)

<u>Marking Guide:</u>

Since the matrix A is already in lower triangular form, the eigenvalues are $\lambda = 0$ and $\lambda = 1$.

$$A - 0I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 1 \end{pmatrix}$$

$$(A - 0I)^2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & -2 & 1 \end{pmatrix}$$

$$(A-0I)^3 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ -1 & -2 & -1 & 0 & -3 & 1 \end{pmatrix}$$

$$(A-0I)^4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ -2 & -3 & -1 & 0 & -4 & 1 \end{pmatrix}$$

$$(A-0I)^5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ -3 & -4 & -1 & 0 & -5 & 1 \end{pmatrix}$$

We have

$r_0(0) = 6$

$r_1(0) = \text{rank}(A-0I) = 4$

$r_2(0) = \text{rank}((A-0I)^2) = 3$

$r_3(0) = \text{rank}((A-0I)^3) = 2$

$r_4(0) = \text{rank}((A-0I)^4) = 2$

$r_5(0) = \text{rank}((A-0I)^5) = 2$

Hence, for the eigenvalue $\lambda = 0$,

Number of Jordan blocks of size 1 is $r_0(0) - 2r_1(0) + r_2(0) = 1$

Number of Jordan blocks of size 2 is $r_1(0) - 2r_2(0) + r_3(0) = 0$

Number of Jordan blocks of size 3 is $r_2(0) - 2r_3(0) + r_4(0) = 1$

Number of Jordan blocks of size 4 is $r_3(0) - 2r_4(0) + r_5(0) = 0$

$$A - I = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & -1 & 0 \end{pmatrix}$$

$$(A-I)^2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 0 & 0 \\ -3 & 2 & -2 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

$$(A-I)^3 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ -3 & 0 & -1 & 0 & 0 & 0 \\ 6 & -3 & 3 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 0 & 0 & 0 \end{pmatrix}$$

We have

$r_0(1) = 6$
$r_1(1) = \text{rank}(A-I) = 5$
$r_2(1) = \text{rank}((A-I)^2) = 4$
$r_3(1) = \text{rank}((A-I)^3) = 4$
Hence, for the eigenvalue $\lambda = 1$,
Number of Jordan blocks of size 1 is $r_0(1) - 2r_1(1) + r_2(1) = 0$
Number of Jordan blocks of size 2 is $r_1(1) - 2r_2(1) + r_3(1) = 1$
It follows that the Jordan canonical form of A is

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

Question 2 (Total: 15 marks)

Let

$$w_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 2 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}, w_3 = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

be linearly independent vectors in $\mathbb{R}^4$. Define $\mathcal{B} = \{w_1, w_2, w_3\}$ and let W be the subspace of $\mathbb{R}^4$ spanned by the ordered basis $\mathcal{B}$.

(a) Use the Gram-Schmidt orthonormalization process to derive an orthonormal basis $\mathcal{C}$ of W.

(7 marks)

(b) A linear operator $T: W \to W$ is defined by

$$T(w_1) = w_1 + 3w_2$$

$$T(w_2) = w_1 - w_3$$

$$T(w_3) = 2w_2 - w_3.$$

Let T^* denote the adjoint of T with respect to the standard inner product. Compute $[T^*]_{\mathcal{C}}$, the matrix representation of T^* relative to the orthonormal basis $\mathcal{C}$ derived in (b).

(8 marks)

<u>Marking Guide:</u>

(a)

Let $u_1 = w_1 = (0,2,1,2)^T$.

Let $u_2 = w_2 - \dfrac{\langle w_2, u_1 \rangle}{\| u_1 \|^2} \cdot u_1$

$$= (0,1,0,2)^T - \frac{\left\langle (0,1,0,2)^T, (0,2,1,2)^T \right\rangle}{\left\| (0,2,1,2)^T \right\|^2} \cdot (0,2,1,2)^T$$

$$= (0,1,0,2)^T - \frac{6}{9} \cdot (0,2,1,2)^T$$

$$= (0,-1/3,-2/3,2/3)^T.$$

Let $u_3 = w_3 - \frac{\langle w_3, u_1 \rangle}{\| u_1 \|^2} \cdot u_1 - \frac{\langle w_3, u_2 \rangle}{\| u_2 \|^2} \cdot u_2$

$$= (-1,1,-1,0)^T - \frac{\left\langle (-1,1,-1,0)^T, (0,2,1,2)^T \right\rangle}{\left\| (0,2,1,2)^T \right\|^2} \cdot (0,2,1,2)^T$$

$$- \frac{\left\langle (-1,1,-1,0)^T, (0,-1/3,-2/3,2/3)^T \right\rangle}{\left\| (0,-1/3,-2/3,2/3)^T \right\|^2} \cdot (0,-1/3,-2/3,2/3)^T$$

$$= (-1,1,-1,0)^T - \frac{1}{9} \cdot (0,2,1,2)^T - \frac{1/3}{1} \cdot (0,-1/3,-2/3,2/3)^T$$

$$= (-1,8/9,-8/9,-4/9)^T$$

Therefore,

$$\left\{ \frac{1}{\sqrt{9}} (0,2,1,2)^T, \frac{1}{\sqrt{1}} \left(0,-\frac{1}{3},-\frac{2}{3},\frac{2}{3}\right)^T, \frac{1}{\sqrt{25/9}} \left(-1,\frac{8}{9},-\frac{8}{9},-\frac{4}{9}\right)^T \right\}$$

$$= \left\{ \left(0,\frac{2}{3},\frac{1}{3},\frac{2}{3}\right)^T, \left(0,-\frac{1}{3},-\frac{2}{3},\frac{2}{3}\right)^T, \left(-\frac{3}{5},\frac{8}{15},-\frac{8}{15},-\frac{4}{15}\right)^T \right\}$$

is an orthonormal basis of W.

(b)

$$[T]_{\mathrm{B}} = \begin{pmatrix} 1 & 1 & 0 \\ 3 & 0 & 2 \\ 0 & -1 & -1 \end{pmatrix}.$$

From the Gram-Schmidt process executed in (a),

$u_1 = w_1$

$u_2 = -\frac{6}{9} w_1 + w_2$

$u_3 = -\frac{1}{9} w_1 - \frac{1}{3} u_2 + w_3 = -\frac{1}{9} w_1 - \frac{1}{3}(-\frac{6}{9} w_1 + w_2) + w_3 = \frac{1}{9} w_1 - \frac{1}{3} w_2 + w_3$.

The orthonormal basis C derived in (a) is given by $\mathrm{C} = \{v_1, v_2, v_3\}$, where

$$v_1 = \frac{1}{3}u_1 = \frac{1}{3}w_1$$

$$v_2 = u_2 = -\frac{6}{9}w_1 + w_2$$

$$v_3 = \frac{3}{5}u_3 = \frac{1}{15}w_1 - \frac{1}{5}w_2 + \frac{3}{5}w_3$$

Let $M_{\mathrm{C,B}}$ denote the transition matrix (change of basis matrix) from C to B.

Then $M_{\mathrm{C,B}} = \begin{pmatrix} 1/3 & -6/9 & 1/15 \\ 0 & 1 & -1/5 \\ 0 & 0 & 3/5 \end{pmatrix}$.

We have $M_{\mathrm{B,C}} = (M_{\mathrm{C,B}})^{-1} = \begin{pmatrix} 3 & 2 & 1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 5/3 \end{pmatrix}$.

Hence,

$$[T]_{\mathrm{C}} = M_{\mathrm{B,C}} \cdot [T]_{\mathrm{B}} \cdot M_{\mathrm{C,B}}$$

$$= \begin{pmatrix} 3 & 2 & 1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 5/3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 3 & 0 & 2 \\ 0 & -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1/3 & -6/9 & 1/15 \\ 0 & 1 & -1/5 \\ 0 & 0 & 3/5 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & -10/3 & 34/15 \\ 1 & -7/3 & 19/15 \\ 0 & -5/3 & -2/3 \end{pmatrix}$$

It follows that

$$[T^*]_{\mathrm{C}} = ([T]_{\mathrm{C}})^T = \begin{pmatrix} 3 & 1 & 0 \\ -10/3 & -7/3 & -5/3 \\ 34/15 & 19/15 & -2/3 \end{pmatrix}.$$

Question 3 (Total: 15 marks)

Let

$$A = \begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 2 & -1 \end{pmatrix},$$

and let $B = A^*A$.

(a) Prove that $B = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$ and that the eigenvalues of B are 3 and 9.

(3 marks)

(b) Demonstrate the property of normal operators by deriving a unitary matrix Q and a diagonal matrix D such that $Q^*BQ = D$.

(6 marks)

(c) Provide unitary matrices U, V and a matrix Σ with non-negative entries satisfying $\text{rank}(\Sigma) = \text{rank}(A)$ such that $A = U\Sigma V^*$ is the singular value decomposition of A.

(6 marks)

<u>Marking Guide:</u>

(a)

$$B = A^*A = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 2 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}.$$

The characteristic equation is

$$\det(xI - B) = \det\begin{pmatrix} x-6 & 3 \\ 3 & x-6 \end{pmatrix}$$

$$= (x-6)^2 - 9 = (x-3)(x-9).$$

Hence, the eigenvalues which are given by solving $(x-3)(x-9) = 0$ are 3 and 9.

(b)

For the eigenvalue $\lambda = 3$,

we solve $\begin{pmatrix} -3 & 3 \\ 3 & -3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

to obtain an eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For the eigenvalue $\lambda = 9$,

we solve $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

to obtain an eigenvector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

These two eigenvectors are already orthogonal.

We normalize each eigenvector and set $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

Also, let $D = \begin{pmatrix} 9 & 0 \\ 0 & 3 \end{pmatrix}$.

Then Q is unitary, D is the diagonal matrix of eigenvalues, and $Q^* B Q = D$.

(c)

The square roots of the eigenvalues of B are $\sqrt{9} = 3$ and $\sqrt{3}$.

Let $\Sigma = \begin{pmatrix} 3 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{pmatrix}$.

Let $V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$ (the same as the matrix Q in (b)).

We compute:

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 2 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$= \frac{1}{3\sqrt{2}}\begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

and

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 2 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

A unit vector u_3 orthogonal to both u_1 and u_2 is $u_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$.

Let $U = \begin{pmatrix} 0 & 2/\sqrt{6} & -1\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1\sqrt{3} \end{pmatrix}$.

Then U and V are unitary, and $A = U \Sigma V^*$ is the singular value decomposition of A.

Question 4 (Total: 20 marks)

Let V denote the vector space of polynomials of degree at most 2 over $\mathbb{R}$. Let

$$\mathcal{B} = \{1 + x, x + x^2, 1 + x + x^2\}$$

be a basis of V, and let $\mathcal{B}^* = \{f_1, f_2, f_3\}$ be the dual basis of $\mathcal{B}$.

(a) Find a matrix $D = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{pmatrix}$ such that

$$f_1(a + bx + cx^2) = d_{11}a + d_{12}b + d_{13}c$$

$$f_2(a + bx + cx^2) = d_{21}a + d_{22}b + d_{23}c$$

$$f_3(a + bx + cx^2) = d_{31}a + d_{32}b + d_{33}c$$

for every polynomial $a + bx + cx^2 \in V$.

(5 marks)

(b) Let W be the subspace of V spanned by the polynomial $1 + 2x + 3x^2$, and let W^0 be the subspace of V^* that annihilates W. If $\mathcal{C} = \{g_1, g_2\}$ is a basis of W^0, express g_1 and g_2 as linear combinations of f_1, f_2, f_3.

(8 marks)

(c) Let h_1, h_2, h_3 be linear functionals in V^* satisfying

$$h_1(p(x)) = p(-1),\ h_2(p(x)) = p(0),\ h_3(p(x)) = p(1).$$

for any polynomial $p(x) \in V$. Prove that $\mathcal{D}^* = \{h_1, h_2, h_3\}$ is a basis of V^* and find a basis $\mathcal{D}$ of V such that $\mathcal{D}^*$ is the dual basis of $\mathcal{D}$.

(7 marks)

Marking Guide:

(a)

The coordinate vectors of $1+x$, $x+x^2$ and $1+x+x^2$ in terms of the standard basis of V are given by

$\begin{pmatrix}1 & 1 & 0\end{pmatrix}^T$

$\begin{pmatrix}0 & 1 & 1\end{pmatrix}^T$

$\begin{pmatrix}1 & 1 & 1\end{pmatrix}^T$

respectively.

Let $A = \begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 1 & 1\end{pmatrix}$.

Then the required matrix is D is given by

$$D = (A^{-1})^T = \begin{pmatrix}0 & 1 & -1\\ -1 & 1 & 0\\ 1 & -1 & 1\end{pmatrix}$$

(b)

We solve for the nullspace of the matrix $\begin{pmatrix}1 & 2 & 3\end{pmatrix}$.

Let $x_2 = s$, $x_3 = t$. Then $x_1 = -2s - 3t$, so

$(x_1, x_2, x_3)^T = (-2s-3t, s, t)^T = s(-2,1,0)^T + t(-3,0,1)^T$.

Hence, a basis of the nullspace is given by

$\{(-2,1,0)^T, (-3,0,1)^T\}$.

We now express the vectors $(-2,1,0)^T, (-3,0,1)^T$ as linear combinations of the rows of D.

By employing Gaussian Elimination, we obtain:

$$\begin{pmatrix} 0 & -1 & 1 & -2 & -3 \\ 1 & 1 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & -2 & -3 \\ -1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 0 & -1 & 1 & -2 & -3 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & 3 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & 3 \\ 0 & 0 & 1 & -1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 & -1 & -3 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 & -2 \end{pmatrix}$$

Thus, we have

$$g_1 = -f_1 + f_2 - f_3$$

and

$$g_2 = -3f_1 + f_2 - 2f_3$$

(c)

Let $h = ah_1 + bh_2 + ch_3$ for real scalars a, b, c, and we show that the equation $h = 0$ only has the trivial solution $a = b = c = 0$.

Applying the linear functional h to the polynomials $1, x, x^2$ respectively, we obtain

$$h(1) = a + b + c$$

$$h(x) = -a + c$$

$$h(x^2) = a + c$$

If $h = 0$, then we obtain the homogenous equation

$$\begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

which clearly only has the trivial solution.

Hence, we conclude that $\{h_1, h_2, h_3\}$ is a basis of V^*.

A dual basis is given by the Lagrange polynomials

$$p_1(x) = \frac{(x-0)(x-1)}{(-1-0)(-1-1)} = \frac{1}{2}(x)(x-1)$$

$$p_2(x) = \frac{(x-(-1))(x-1)}{(0-(-1))(0-1)} = -(x+1)(x-1)$$

$$p_3(x) = \frac{(x-(-1))(x-0)}{(1-(-1))(1-0)} = \frac{1}{2}(x+1)(x)$$

Question 5 (Total: 10 marks)

Let V be a complex inner product space and let $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ be an orthonormal basis of V.

(a) Prove that for any $x \in V$,

$$x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i.$$

(5 marks)

(b) For any $x, y \in V$, prove that

$$\langle x, y \rangle = \sum_{i=1}^{n} \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

(5 marks)

<u>Marking Guide:</u>

(a)

We have

$$\langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}.$$

For any given $x \in V$, write $x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$.

Then for any $1 \leq j \leq n$, we have

$$\langle x, v_j \rangle = \left\langle \sum_{i=1}^{n} \alpha_i v_i, v_j \right\rangle$$

$$= \sum_{i=1}^{n} \alpha_i \langle v_i, v_j \rangle = \alpha_j$$

It follows that $x = \sum_{i=1}^{n} \langle x, v_i \rangle v_i$.

(b)

For any given $x, y \in V$, we have

$$\langle x, y \rangle = \left\langle \sum_{i=1}^{n} \langle x, v_i \rangle v_i, \sum_{j=1}^{n} \langle y, v_j \rangle v_j \right\rangle$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} \langle \langle x, v_i \rangle v_i, \langle y, v_j \rangle v_j \rangle$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} \langle x, v_i \rangle \overline{\langle y, v_j \rangle} \langle v_i, v_j \rangle$$

$$= \sum_{i=1}^{n} \langle x, v_i \rangle \overline{\langle y, v_i \rangle}.$$

Question 6 (Total: 10 marks)

Let

$$A = \begin{pmatrix} -1 & 3 & 2 \\ 3 & 1 & 4 \\ 2 & 4 & 1 \end{pmatrix}$$

be the matrix representation of a symmetric bilinear form f on $\mathbb{R}^3$ relative to the standard basis. Find an invertible matrix Q such that QAQ^T is a diagonal matrix, and determine the rank and signature of f.

(10 marks)

<u>Marking Guide:</u>

Perform simultaneous row and column operations starting from an augmented matrix in which A is placed on the left and the 3×3 identity matrix is placed on the right:

$$\begin{pmatrix} -1 & 3 & 2 & 1 & 0 & 0 \\ 3 & 1 & 4 & 0 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 3 & 2 & 1 & 0 & 0 \\ 0 & 10 & 10 & 3 & 1 & 0 \\ 2 & 4 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 10 & 10 & 3 & 1 & 0 \\ 2 & 10 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 10 & 10 & 3 & 1 & 0 \\ 0 & 10 & 5 & 2 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 10 & 10 & 3 & 1 & 0 \\ 0 & 10 & 5 & 2 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 10 & 10 & 3 & 1 & 0 \\ 0 & 0 & -5 & -1 & -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 10 & 0 & 3 & 1 & 0 \\ 0 & 0 & -5 & -1 & -1 & 1 \end{pmatrix}$$

Let $Q = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$.

Then Q is an invertible matrix such that $QAQ^T = \text{diag}(-1, 10, -5)$.

The rank of f is 3.

The signature of f is $(1, 2, 0)$.

Also accept the answer: The signature of f is $1 - 2 = -1$ (as these are the two common definitions of signature of a symmetric bilinear form over $\mathbb{R}$).

Question 7 (Total: 15 marks)

Let A be a complex $n \times n$ matrix, and let A^* denote the conjugate transpose of A.

(a) Show that the nullspace of A^*A is equal to the nullspace of A.

(6 marks)

(b) If $A^*A = AA^*$, prove that the nullspace of A^* is equal to the nullspace of A.

(3 marks)

(c) Suppose that A has polar decomposition $A = UP$, where U is unitary, P is self-adjoint and non-negative. Prove that $A^*A = AA^*$ if and only if $UP^2 = P^2U$.

(6 marks)

<u>Marking Guide:</u>

(a)

Let $x \in N(A)$

Then $Ax = 0$, so $A^*Ax = 0$. Thus, $x \in N(A^*A)$.

Hence, $N(A) \subseteq N(A^*A)$.

Let $x \in N(A^*A)$.

Then $A^*Ax = 0$, so $x^*A^*Ax = 0$.

Thus, $\| Ax \|^2 = 0$, so $Ax = 0$.

Hence, $N(A^*A) \subseteq N(A)$.

We deduce that $N(A^*A) = N(A)$.

(b)

Suppose that $A^*A = AA^*$.

By (a), $N(A^*A) = N(A)$, and

$N(AA^*) = N(A^*)$.

Since $A^*A = AA^*$, we deduce that $N(A) = N(A^*)$.

(c)

We have $A^*A = (UP)^*(UP) = P^*U^*UP = P^*P = P^2$.

We also have $AA^* = (UP)(UP)^* = UPP^*U^* = UP^2U^*$

If $A^*A = AA^*$, then $P^2 = UP^2U^*$, and so multiplying on the right by U yields $P^2U = UP^2$.

If $P^2U = UP^2$, then multiplying on the right by U^* yields $P^2 = UP^2U^*$, and so $A^*A = AA^*$.

----- END OF MARKING GUIDE -----